APPL: A Prompt Programming Language for Harmonious Integration of Programs and Large Language Model Prompts

Honghua Dong[1,2,*], **Qidong Su**[1,2,3,*], **Yubo Gao**[1,2,3], **Zhaoyu Li**[1,2], **Yangjun Ruan**[1,2], **Gennady Pekhimenko**[1,2,3,4], **Chris J. Maddison**[1,2,4], **Xujie Si**[1,2,4]

[1]University of Toronto [2]Vector Institute [3]CentML [4]CIFAR AI Chair

*Equal contribution

Correspondence: honghuad@cs.toronto.edu, six@cs.torontu.edu

Abstract

Large Language Models (LLMs) have become increasingly capable of handling diverse tasks with the aid of well-crafted prompts and integration of external tools, but as task complexity rises, the workflow involving LLMs can be complicated and thus challenging to implement and maintain. To address this challenge, we propose APPL, A Prompt Programming Language that acts as a bridge between computer programs and LLMs, allowing seamless embedding of prompts into Python functions, and vice versa. APPL provides an intuitive and Python-native syntax, an efficient parallelized runtime with asynchronous semantics, and a tracing module supporting effective failure diagnosis and replaying without extra costs. We demonstrate that APPL programs are intuitive, concise, and efficient through representative scenarios including Chain-of-Thought with self-consistency (CoT-SC) and ReAct tool-use agent. We further use LLMs to judge the language design between APPL and previous work, where the results indicate that codes written in APPL are more readable and intuitive. Our code, tutorial and documentation are available at https://github.com/appl-team/appl.

1 Introduction

Large Language Models (LLMs) have demonstrated remarkable capabilities in understanding and generating texts across a broad spectrum of tasks (Raffel et al., 2020; Brown et al., 2020; Chowdhery et al., 2023; Touvron et al., 2023a,b; OpenAI, 2023a). Their success has contributed to an emerging trend of regarding LLMs as novel operating systems (Andrej Karpathy, 2023; Ge et al., 2023; Packer et al., 2023). Compared with traditional systems that precisely execute structured and well-defined programs written in programming languages, LLMs can be guided through flexible natural language prompts to perform tasks that are beyond the reach of traditional programs.

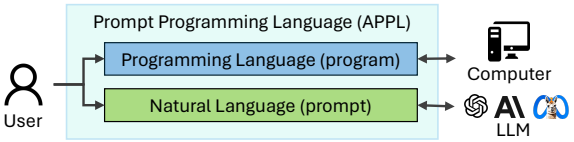


Figure 1: We introduce APPL, a *prompt programming language* that integrates conventional programs and natural language prompts to provide a unified interface for users to access computers and LLMs together. APPL also facilitates users fusing the strengths of computer programs and LLMs by providing convenient conventions between them.

Meanwhile, there is a growing interest in harnessing the combined strengths of LLMs and conventional computing to address more complex challenges. Approaches like tree-of-thoughts (Yao et al., 2024), RAP (Hao et al., 2023) and LATS (Zhou et al., 2023) integrate search-based algorithms with LLM outputs, showing improved outcomes over using LLMs independently. Similarly, the creation of semi-autonomous agents such as AutoGPT (Richards, 2023) and Voyager (Wang et al., 2023) demonstrates the potential of LLMs to engage with external tools, like interpreters, file systems, and skill libraries, pushing the boundaries of what integrated systems can achieve. However, as task complexity rises, the workflow involving LLMs becomes more intricate, requiring more complex prompts guiding LLMs and more sophisticated programs implementing the workflow, which are both challenging to implement and maintain.

To address these challenges, as illustrated in Figure 1, we propose APPL, A Prompt Programming Language for harmonious integration of conventional programming and LLMs. More specifically, APPL's key features include:

(1) Readability and maintainability via seamless integration with Python. As shown in Figure 2, APPL seamlessly embeds natural language prompts to Python programs, maintaining prompts'

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```python
@ppl(ctx="copy") # copy the context from caller
def get_answer(question):
  question # append to the prompt
  return gen() # return the string response
@ppl # marks APPL function
def answer_questions(quotation, questions):
  "Extract the name of the author from the quotation below and answer questions."
  quotation # append to the prompt
  with AIRole(): # assistant message scope
    f"The name of the author is {gen()}"
  return [get_answer(q) for q in questions] # parallelize calls
```

(a) The APPL program for answering questions.

```python
def get_answer(messages, question):
  return gen(messages + [user(question)]) # new message list with addon message
def answer_questions(quotation, questions):
  messages = [user(f"Extract the name of the author from the quotation below and answer questions.\n{quotation}")]
  messages.append(assistant("The name of the author is "))
  messages.append(assistant(gen(messages)))
  return [get_answer(messages, q) for q in questions] # functions called sequentially, need to wait for results
```

(b) The Python program with helper functions.

```python
quotation = '"Simplicity is the ultimate sophistication." -- Leonardo da Vinci'
questions = ["In what era did the author live?", ...]
answer_questions(quotation, questions)
```

(c) Example call to `answer_questions`.

```
User: Extract the name of the author from the quotation below and answer questions.
"Simplicity is the ultimate sophistication." -- Leonardo da Vinci
Assistant: The name of the author is Leonardo da Vinci.
User: In what era did the author live?
Assistant: Leonardo da Vinci lived during the Renaissance era.
```

(d) Example LLM prompts and responses for the first question.

Figure 2: **(a)** The APPL program for answering multiple questions about a quotation by first extracting the author's name. In this APPL program, the `@ppl` marks the function to be APPL function, which provides a *context* that interactive expression statements (highlighted lines) can be interacted with. Passing the context across APPL functions is implicitly handled along with function calls, mimicking the process of stack frame management. The `gen` function requests an LLM generation using the prompts stored in the context, with other optional arguments following the OpenAI's API (OpenAI). For instance, the `gen` in line 4 uses the prompt composed by (highlighted) lines 7,8,10 (since context is copied), and 3. The LLM generation is computed asynchronously and only synchronizes when the value is needed, therefore independent calls are automatically parallelized as shown in line 11. **(b)** A Python program implementing the same function uses more codes, even with the helper functions. Furthermore, extra codes are needed to parallelize `get_answer` function calls. **(c)** Example usage for the program and **(d)** its corresponding LLM prompts and example responses.

readability while inheriting modularity, reusability, and dynamism from the host language.

(2) Automatic parallelization via asynchronous computation. APPL schedules LLM calls asynchronously, leveraging potential independence among them to facilitate parallelization. This offloads the burden of users to manage synchronization manually, with almost no extra work as shown in Figure 2a. Its effectiveness is validated in Section 5.2 where experiments show significant accelerations for three popular applications.

(3) Smooth transition between structured data and natural language. APPL enables seamlessly converting program objects into prompts (namely `promptify`). For example, `promptify`-ing Python functions converts them into LLM-understandable tool specifications. Specifically, APPL implements an `as_tool` function that analyzes a Python function's docstrings and signatures to create tool specifications (in JSON schema format) that being used by LLM calls. In another direction, APPL enables the specification of output structures for LLMs, such as ensuring outputs conform to Pydantic's BaseModel (Colvin et al.).

APPL strives to be a powerful, yet intuitive extension of Python and leverages the unique capabilities of LLMs while being compatible with Python's syntax and ecosystem. It empowers developers to easily build sophisticated programs that harness the power of LLMs in a manner that's both efficient and accessible.

2 Related Work

LLM programs. The programs involving LLMs continue evolving sophisticated to tackle more challenging tasks, from few-shot prompts (Brown et al., 2020) to elaborated thought process (Wei et al., 2022; Ning et al., 2023; Yao et al., 2024; Besta et al., 2023), from chatbot (OpenAI, 2022) to tool-use agents (Nakano et al., 2021; Ahn et al., 2022; Liu et al., 2022; Liang et al., 2023; Richards, 2023; Patil et al., 2023; Wang et al., 2023; Qin et al., 2023), virtual emulators (Ruan et al., 2024), operating systems (Packer et al., 2023), and multi-agent systems (Park et al., 2023; Hong et al., 2023; Qian et al., 2023). The increasing complexity necessitates powerful frameworks to facilitate the development of those programs.

LLM software stack. On the low level, LLM services are provided by high-performance back-

Features	LMQL (Beurer-Kellner et al., 2023)	Guidance (GuidanceAI, 2023)	SGLang (Zheng et al., 2023)	**APPL**
Language Syntax	Python-like	Python	Python	Python
Parallelization and Asynchrony	Sub-optimal	Manual	**Auto**	**Auto**
Prompt Capturing	**Auto**	Manual	Manual	**Auto**
Prompt Context Passing	Limited	Manual	Manual	**Auto**
Promptify Tool	Manual	Manual	Manual	**Auto**
Prompt Compositor	No	No	No	**Yes**
Generation Output Retrieval	Template variables	`str`-index	`str`-index	**Python-native**
Generation without Side-effects	No	No	No	**Yes**
Failure Recovery	No	No	No	**Yes**

Table 1: Comparison across Prompt Languages. APPL enhances the user experience of crafting LLM-augmented programs with Python-native syntax while achieving high performance with an efficient parallelizable runtime system. We carefully design the language features involved in prompting and generation, which streamlines the whole workflow. APPL also provides failure recovery through its tracing mechanism. See Sections 3 for the language design and Section 5 for more comparison details.

ends (Gerganov, 2023; HuggingFace, 2023; Kwon et al., 2023) as APIs like OpenAI APIs (OpenAI). Building on this, modules with specific functionality are invented, like unifying APIs (BerriAI, 2023), constraining LLM outputs (Willard and Louf, 2023; Liu, 2023) and parallelizing function calls (Kim et al., 2023), where APPL utilizes some of them as building blocks. LMQL (Beurer-Kellner et al., 2023), Guidance (GuidanceAI, 2023), and SGLang (Zheng et al., 2023) are languages similar to APPL. These languages provide developers with precise tools for crafting sophisticated model interactions, enabling finer-grained control over LLM behaviors. We compare APPL with them by case studies in Section 5 and summarize the comparison in Table 1, where APPL provides more automatic supports for prompting and parallelizing LLM calls to streamline the whole workflow. There are also higher-level general-purpose frameworks (Okuda and Amarasinghe, 2024; Prefect, 2023; Khattab et al., 2023; Chase, 2022) and agent-specific frameworks (Team, 2023; Li et al., 2023; Wu et al., 2023). The pipelines in these frameworks can also be implemented using prompt languages like APPL.

3 Language Design and Features

3.1 Design Principles of APPL

(1) Readability and Flexibility. Natural language provides a readable and understandable human-LLM interface, while programming languages enable flexible controls, clear structures, modularization, reusability, and external tools. APPL aims to utilize both strengths.

(2) Easy Parallelization. Parallelization is a critical performance optimization for LLM text generation (Yu et al., 2022; Kwon et al., 2023). APPL aims to provide transparent support for parallelization with almost no extra code introduced.

(3) Convenient Conversion. APPL aims to facilitate easy conversion between structured data (including codes) for programming and the natural language for LLMs.

3.2 APPL Syntax and Semantics

As illustrated in the APPL program of Figure 2a, the syntax and semantics of APPL are extended from Python with slight modifications to integrate natural language prompts and LLMs calls into the traditional programming constructs deeply.

APPL functions are the fundamental building blocks of APPL, denoted by the `@ppl` decorator. Each APPL function has an implicit context to store prompt-related information. Within APPL functions, the expression statements are interpreted as "interactions" with the context, where the typical behavior is to append strings to the prompt.

Expression statements are standalone Python statements formed by an expression, for example, `"str"`, while an assignment (e.g., `a = "str"`) is not. Their evaluation result is handled differently according to the environment. For example, it causes no extra effect in standard Python while the result is displayed in the interactive environment in Jupyter. Similarly, APPL provides an environment where each APPL function contains a scratchpad of prompts (i.e. context), and the values of expression statements will be converted to prompts (via `promptify`) and appended to the prompt scratchpad. For example, a standalone string as line 7 or 8 in Figure 2a will be appended to the scratchpad.

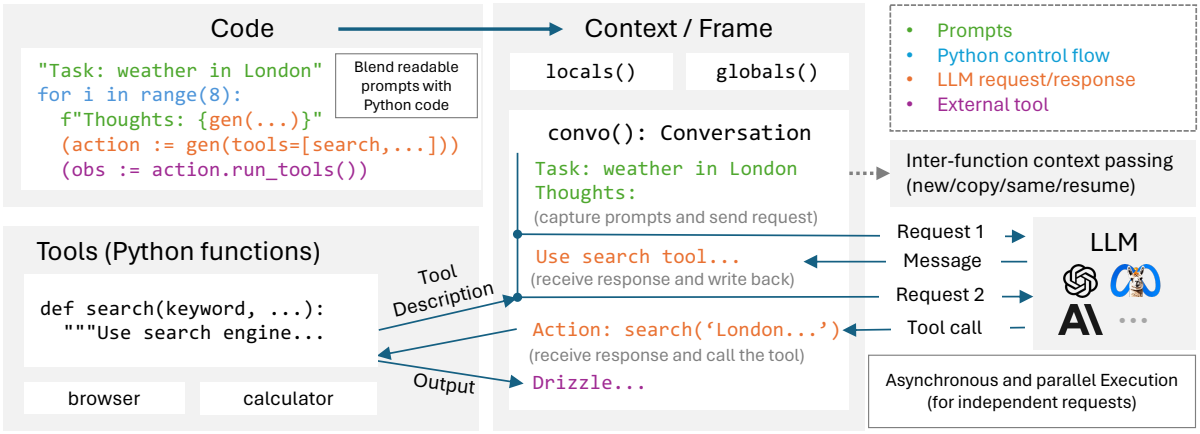


Figure 3: Overview of the APPL runtime. The code is executed following Python control flows, with prompts and LLM responses being captured into the `conversation` (can be retrieved by `convo()`, similar to `locals()` and `globals()` in the Python frame). The gen uses the `conversation` captured so far as prompts to call LLMs and is executed asynchronously, so that independent calls can be parallelized easily. APPL allows using Python functions as tools for LLMs by extracting information from their docstrings and signatures. When tools are provided, their specifications are included in the request and the returned tool calls can be easily executed to get results. See detailed code in Figure 9a.

```
1  Req = define("Requirement")
2  InputReq = define(f"Input {Req}")
3  OutputReq = define(f"Output {Req}")
4  @ppl
5  def func():
6    f"... the following {Req}s:"
7    with NumberedList(indent=2):
8      InputReq(desc="Input should ...")
9      OutputReq(desc="Output should ...")
10   return records()
```

```
1  >>> print(func())
2  ... the following Requirements:
3    1. Input Requirement: Input should ...
4    2. Output Requirement: Output should ...
```

Figure 4: Illustration of the usage of *Definitions* and *Prompt Compositors*.

LLM generations (gen), when being called, implicitly use the accumulated prompt in the context of the APPL function. A special case is the standalone f-string, which is split into parts and they are captured as prompts in order, detailed in Appendix A.1. This provides a more readable way to combine prompts and LLM generations in one line. For example, line 10 of Figure 2a means start generation by first adding the prefix "The name of the author is " to the prompt, which matches human intuition.

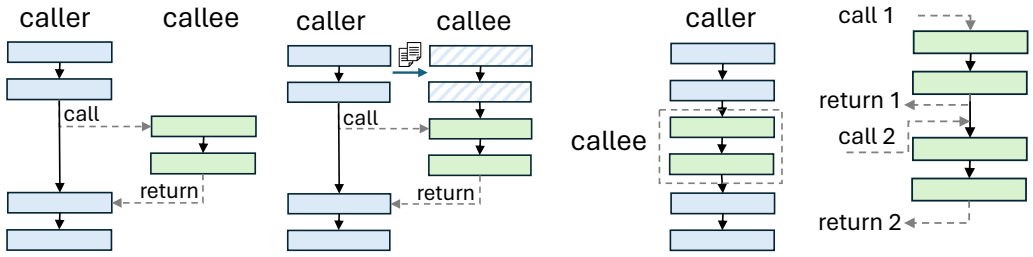
Context managers modify how prompt statements are accumulated to the prompt in the context. APPL provides two types of context managers:
(1) *Role Changers* (such as `AIRole` in Figure 2a) are used in chats to specify the owner of the message, and can only be used as the outmost one.
(2) *Prompt Compositors* (such as `NumberedList` in Figure 4) specify the delimiter, indention, indexing format, prologue, and epilogue. These compositors allow for meticulously structured finer-grained prompts in hierarchical formatting through native Python syntax, greatly enhancing flexibility without losing readability.

Definition is a special base class for custom definitions. As shown in Figure 4, once declared, *Definitions* can be easily referred to in the f-string, or used to create an instance by completing its description. They are useful for representing concepts that occur frequently in the prompt and support varying the instantiation in different prompts. See more usage in Appendix G when building a long prompt.

3.3 APPL Runtime

APPL runtime context. Similar to Python's runtime context that contains local and global variables (`locals()` and `globals()`), the context of APPL functions contains local and global prompts that can be accessed via `records()` and `convo()`. The example in Figure 3 illustrates how the prompts in the context changed during the execution.

Context passing across functions. When a APPL function (caller) calls another APPL function (callee), users might expect different behaviors for initializing the callee's context depending on specific use cases. APPL provides four ways as shown in Figure 5:



(a) new (b) copy (c) same (d) resume

Figure 5: Illustration of four different ways of inter-function context passing. Usage examples can be found in Section E and Appendix A.2. Prompts connected by solid arrows belong to the same prompt context. Dashed arrows represent the control flow of function calls and returns.

1. `new`: Creates a clean context. This is the default and most common way.

2. `copy`: Creates a copy of the caller's context. This is similar to "call by value" where the modification to the copy will not influence the original context. This is useful for creating independent and asynchronize LLM calls as shown in Figure 2a.

3. `same`: Uses the same context as the caller. This is similar to "call by reference" where the modifications are directly applied to the original context.

4. `resume`: Resumes callee's context from the last run. This makes the function stateful and suitable for building interactive agents with history (see Figure 18b for an example).

When decomposing a long prompt into smaller modules, both `new` and `same` can be used as shown in Figure 11, but the default way (`new`) should be prioritized when possible.

Asynchronous semantics. To automatically leverage potential independence among LLM calls and achieve efficient parallelization, we borrow the idea of the asynchronous semantics from PyTorch (Paszke et al., 2019). Asynchronous semantics means the system can initiate operations that run independently of the main execution thread, allowing for parallelism and often improving performance. In APPL, LLM calls (`gen`) are asynchronous, which means the main thread proceeds without being blocked by `gen` until accessing the generation results is necessary. See implementation details in Section 4.2.

Tool calling. Tool calling is an indispensable ability for LLM agents to interact with external resources. To streamline the process of providing tools to LLMs, APPL automatically creates

```python
def is_lucky(x: int) -> bool:
  """Determine whether the input number is a lucky number.
  <Args and Returns omitted ... >"""
  return sympy.isprime(x+3)
@ppl
def answer_with_tool(question: str):
  question # E.g., Is 2024 a lucky number?
  # Generation with tool integration
  action = gen(tools=[is_lucky, search])
  # Execute the tools and get the results
  results = action.run_tool_calls()
  return results[0].content
```

Figure 6: Tool calling example in APPL.

tool specifications from Python functions by analyzing their docstrings and signatures (detailed in Appendix A.3). As shown in Figure 6, the Python functions `is_lucky` and `search` can be directly utilized as tools for LLMs. Therefore, thanks to the well-written docstrings, the functions from Python's rich ecosystem can be integrated into LLMs as tools with minimal effort. Moreover, APPL automatically converts LLMs' tool call outputs as function calls that can be directly executed to obtain results.

Tracing and caching. Besides standard caching for LLM calls, APPL supports tracing APPL functions and LLM calls to facilitate users to understand and debug the program executions. The trace is useful for reproducing (potentially partial) execution results by loading cached responses of the LLM calls, which enables failure recovery and avoids the extra costs of resending these calls. This also unlocks the possibility of debugging one LLM call conveniently. Users can also visualize the program execution to analyze the logic, expense, and runtime of the program. APPL provides two modes of tracing, namely strict and non-strict modes, corresponding to different reproduction requirements (perfect or statistical reproduction). We further explain the implementation details for handling the asynchronous semantics in Appendix F.1, and demonstrate the trace visualization in Appendix F.2.

4 Implementation

In this section, we elaborate on the implementation details of APPL, including compilation and asynchronous execution.

4.1 Compilation

To provide the runtime context for APPL functions more smoothly and make it transparent to users, we choose to compile (more specifically, transpile) the Python source code to insert the context into the code. We use Python's `ast` package to parse the original function as AST, mainly apply the following code transformations on the AST, and compile back from the modified AST to a Python function. Appendix D gives an example of the code before and after transpilation.

- **Context management.** We use the keyword `_ctx` to represent the context of the function, which is inserted in both the definition of APPL functions and the function calls that need the context. The functions that require the context as inputs are annotated by the attribute `__need_ctx__`, including all functions introduced in Section 3.2: APPL functions, `gen` and context manager functions.
- **Capture expression statements.** We add an execution function as a wrapper for all expression statements within the APPL function and let them "interact" with the context. The execution behavior differs based on the value types, where the behavior for the most common `str` type is appending it to the prompt of the context. For a standalone f-string, we split it into several statements to let them be "executed" in order so that `gen` functions in the f-string can use the text before them. This procedure is detailed in Appendix A.1.

4.2 Asynchronous Execution and Future Objects

To implement the asynchronous semantics, we adopt the concept of `Future` introduced in the Python standard library `concurrent.futures`, which represents the asynchronous execution of a callable. When the result of `Future` is accessed, it will wait for the asynchronous execution to finish and synchronize the result.

We introduce `StringFuture`, which represents a string object, but its content may not be ready yet. `StringFutures` behave almost identically to the normal `str` objects in Python, so that users can use them as normal strings transparently and enjoy the benefits of asynchronous execution. Meanwhile, the `StringFuture` delays its synchronization when possible, ideally only synchronizing when the `str` method is called.

Formally, we define `StringFuture` S to be a list of `StringFutures`: $[s_1, s_2, \ldots, s_n]$. Without loss of generality, we can regard a normal string as an already synchronized `StringFuture`. A `StringFuture` S is synchronized when $\texttt{str}(S)$ is called, and it collapses to a normal string by waiting and concatenating the results of $\texttt{str}(s_i)$:

$$\texttt{str}([s_1, \ldots, s_n]) \rightarrow \texttt{str}(s_1) \oplus \texttt{str}([s_2, \ldots, s_n]),$$

where $\oplus$ operation means (string) concatenation. Such representation enables delaying the concatenation ($\oplus$) of two `StringFutures` S and T as concatenating ($\oplus$) two lists:

$$\begin{aligned} S \oplus T &= [s_1, \ldots, s_n] \oplus [t_1, \ldots, t_m] \\ &= [s_1, \ldots, s_n, t_1, \ldots, t_m]. \end{aligned}$$

For other methods that cannot be delayed, like $\texttt{len}(S)$, it falls back to materializing S to a normal string and then calls the counterpart method.

Similarly, we introduce `BooleanFuture` to represent a boolean value that may not be synchronized yet, which can be used to represent the comparison result between two `StringFutures`. A `BooleanFuture` B is synchronized only when `bool(B)` is called, which enables using future objects in control flows.

5 Case Studies and Language Comparisons

In this section, we compare APPL with other prompt languages including LMQL (Beurer-Kellner et al., 2023), Guidance (GuidanceAI, 2023), and SGLang (Zheng et al., 2023) (whose syntax is mostly derived from Guidance) through usage scenarios, including parallelized LLM calls like the chain of thoughts with self-consistency (CoT-SC) (Wang et al., 2022), tool use agent like ReAct (Yao et al., 2023), and multi-agent chat. We further include a re-implementation of a long prompt (more than 1000 tokens) from ToolEmu (Ruan et al., 2024) using APPL in Appendix G, demonstrating the usage of APPL functions and prompt compositors for modularizing and structuring long prompts.

```python
@ppl
def cot(num_trials: int):
  "... (three examples omitted)"
  "Q: You need to cook 9 eggs."
  "A: Let's think step by step."
  return [gen() # parallel
    for _ in range(num_trials)]

answers = cot(num_trials)
```

(a) APPL

```python
@lmql.query
async def cot():
  '''lmql
  "... (three examples omitted)"
  "Q: You need to cook 9 eggs."
  "A: Let's think step by step.[ANSWER]"
  return ANSWER
  '''

async def cot_parallel(num_trials: int):
  responses = [cot() for i in range(num_trials)]
  return await asyncio.gather(*responses)

answers = asyncio.run(cot_parallel)
```

(b) LMQL

```python
@function
def cot(s, num_trials: int):
  s += "... (three examples omitted)\n"\
       "Q: You need to cook 9 eggs.\n"\
       "A: Let's think step by step."
  forks = s.fork(num_trials)
  for fork in forks:
    fork += gen("answer") # parallel in forks
  return [fork["answer"] for fork in forks]

answers = cot.run(num_trials).ret_value
```

(c) SGLang

```python
def cot(lm, num_trials: int):
  lm += "... (three examples omitted)\n"\
        "Q: You need to cook 9 eggs.\n"\
        "A: Let's think step by step."
  answers = []
  for i in range(num_trials): # sequential
    fork = lm # avoid mutating lm
    fork += gen("answer") # become a new object
    answers.append(fork["answer"])
  return answers

answers = cot(lm, num_trials)
```

(d) Guidance

Figure 7: Comparison of CoT-SC implementations in APPL, LMQL, SGLang, and Guidance, where the marginalizing procedure in CoT-SC is omitted. APPL demonstrates its simplicity and clarity, and exhibits one or more advantages in terms of **1) asynchrony and parallelization**, **2) generation and output retrieval**, and **3) context management**, when comparing with others.

5.1 Chain of Thoughts with Self-Consistency

Compared with CoT-SC implemented in LMQL, SGLang and Guidance, APPL exhibits its conciseness and clarity, as shown in Figure 7. We mainly compare them from the following aspects:

Parallization. APPL performs automatic parallelization while being transparent to the programmer. In particular, since the returned value need not be immediately materialized, the different branches can execute concurrently. SGLang allows for parallelization between subqueries by explicitly calling the `lm.fork()` function. LMQL employs a conservative parallelization strategy, which waits for an `async` function immediately after it's called. To support the same parallelization strategy as in APPL, we manually implement the same parallelization strategy use the `asyncio` library to launch LMQL queries in parallel at the top level. Guidance does not include asynchrony or parallelization as a core feature.

Generation and output retrieval. For LLM generation calls, APPL, SGLang, and Guidance use a Python function while LMQL uses inline variables (with annotations), which is less flexible for custom generation arguments. For the results of generations, both SGLang and Guidance store them as key-value pairs in a dictionary associated with the context object and are indexed with the assigned key when required. LMQL accesses the result by the declared inline variable, but the variable is not Python-native. In APPL, `gen` is an independent function call, whose return value can be assigned to a Python variable, which is *Python-native* and allows for better interoperability with IDEs. Moreover, APPL separates the prompt reading and writing for the generation. Users have the option to write generation results back to the prompt or not (called *without side-effects* in Table 1). This allows a simple replication of LLM generation calls sharing the same prompt. Otherwise, explicit copying or rebuilding of the prompt would be necessary for such requests.

Context management and prompt capturing. Both SGLang and Guidance need to explicitly manage context objects (`s` and `lm`) as highlighted as red in Figure 7. The prompts are captured manually using the `+=` operation. The branching of the context is implemented differently, where SGLang uses the explicit `fork` method and Guidance returns a new context object in the overridden `+=` operation. In contrast, APPL automatically captures standalone expression statements into the prompt context, similar to LMQL which captures standalone strings. The context management is also done automatically for nested function calls in APPL and LMQL, where APPL supports four patterns *new*, *copy*, *same*, and *resume* (see Figures 5 and 2a) while LMQL only supports *new* and *copy*.

Time(s)	CoT-SC (Wang et al., 2022)		SoT (Ning et al., 2023)		MemWalker (Chen et al., 2023)		ToT (Yao et al., 2024)	
	GPT-3.5	LLAMA-7b	GPT-3.5	LLAMA-7b	GPT-3.5	LLAMA-7b	GPT-4o	LLAMA3.2-3B
Sequential	27.6	17.0	227.2	272.3	707.6	120.6	164.3	621
Parallel	2.9	1.8	81.2	85.9	230.3	65.4	17.5	95.9
(estimated speedup)*	10		4.3	3.7	3.8		10	
Speedup	9.49×	9.34×	2.79×	3.17×	3.07×	1.84×	9.39×	6.48×

Table 2: Speedup brought by parallelization with automatic asynchronous execution in APPL. (* *estimated speedup* is calculated using Amdahl's law by assuming the running time of all LLM calls is identical. The number of branches in SoT is determined by LLMs and thus varies for different LLMs. Details about the estimation can be found in Appendix B.1.) MemWalker with LLAMA-7b does not achieve the estimated speedup because its long context lengths increase memory footprint, leading to smaller batch sizes and fewer parallelizable requests.

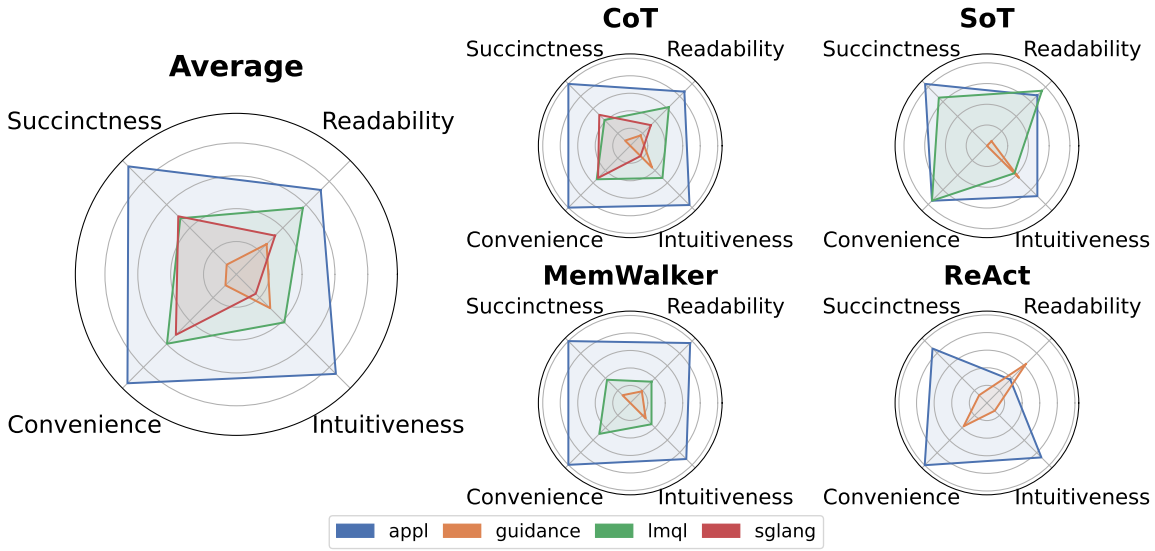


Figure 8: Language model evaluation results in different implementations of algorithms. We use language models (GPT, Claude, DeepSeek) to score various tasks implemented in different prompt languages across multiple aspects.

5.2 Parallelized LLM Calls

We validated the effectiveness of parallelizing LLM calls in APPL in four tasks: question answering with CoT-SC, content generation with skeleton-of-thought (SoT) (Ning et al., 2023),hierarchical summarization with MemWalker (Chen et al., 2023), and Tree of Thoughts (ToT) reasoning (Yao et al., 2024). As shown in Table 2, the parallel version achieves significant speedup across different branching patterns, and the actual speedup ratios almost match the estimated ones. More experimental details can be found in Appendix B.

5.3 ReAct Tool Use Agent

We use the ReAct (Yao et al., 2023) tool-use agent as an example to compare the accessibility of tool calls in different languages. As shown in Figure 9a, the major workflow of ReAct unfolds over several cycles, each comprising three stages: **(1)** generating "thoughts" derived from existing information to steer subsequent actions; **(2)** selecting appropriate actions (tool calls) to execute; **(3)** retrieving the results of executing the actions. These results then serve as observations and influence the subsequent cycle.

LLM Tool calling can be divided into three steps: 1) encoding the external tools as specifications that are understandable by LLMs, 2) parsing generated tool calls, and 3) executing the tool calls. APPL provides out-of-the-box support for these steps based on the OpenAI's API (OpenAI). As shown in Figure 9a, documented Python functions like `is_lucky` defined in Figure 6 can directly be used as the tools argument without manual efforts. Differently, in Figure 9b, Guidance (similarly for SGLang and LMQL) can generate tool calls and parse structured arguments via its `select` and other constraint primitives, but the tool specifications and format constraints need to be written manually. Guidance provides another way to automate this process but with limited support that only works for functions with string-typed arguments.

```
tools = [search, is_lucky, ...]
f"User Instruction: {instruction}"
for i in range(num_iterations):
  with AIRole():
    # Generate thoughts in text given the tools
    f"Thoughts: {gen(tools=tools, tool_choice='none')}"
  # Generate tool calls as actions
  (actions := gen(tools=tools, tool_choice='required'))
  # Run the tool calls to get observations
  (observations := actions.run_tool_calls())
```

(a) The main part of APPL codes to implement the ReAct algorithm.

```
tool_names = ["search", "is_lucky", ...]
tool_map = {"search": search, "is_lucky": is_lucky, ...} # need to rewrite is_lucky to take str input
tool_desc = {"search": "Use search engine...\n parameters:\n  keywords(str): ...\n returns: ...",
             "is_lucky": "Determine whether...\n parameters:\n  x(int): ...\n returns: ...", ...}
lm += f"We have the following tools: {tool_desc}" + f"User Instruction: {instruction}\n"
for i in range(num_iterations):
  lm += f"Thought {i}: " + gen(name="thought", suffix="\n")
  lm += f"Action {i}: " + select(tool_names, name="act") + "(" + gen(name="arg", suffix=")") + "\n"
  lm += f"Observation {i}: " + tool_map[lm["act"]](lm["arg"]) + "\n"
```

(b) The main part of Guidance codes to implement the ReAct algorithm. The Python functions need manual configurations.

Figure 9: Comparison of ReAct implementations in APPL and Guidance. Instead of manually specifying the specifications of tools, APPL automatically creates tool specifications from the signature and docstring of Python functions, making the well-documented Python functions more accessible to LLMs.

	APPL	LMQL		SGLang/Guidance	
Tasks	`AST-size`	`AST-size`	vs. APPL	`AST-size`	vs. APPL
CoT-SC	**35**	57	1.63×	61	1.74×
ReAct	**61**	not support directly		134	2.20×
SoT	**59**	120	2.03×	107	1.81×
MemWalk	**36**	64	1.78×	60	1.67×

Table 3: The number of AST nodes of the programs implemented in different prompting languages across different tasks. Note that LMQL programs need to include extra `asyncio` codes to parallelize LLM calls.

5.4 Code Succinctness Comparison

As demonstrated in Figure 7, APPL is more concise than other compared languages to implement the Cot-SC algorithm. Quantitatively, we further introduce a new metric `AST-size` to indicate the succinctness of programs written in different languages, which counts the number of nodes in the abstract syntax tree (AST) of the code. For Python programs, we can obtain their AST using the standard library with `ast.parse`.

As shown in Table 3, LMQL, SGLang, and Guidance need about twice AST nodes than APPL for implementing the same tasks, highlighting the succinctness of APPL.

5.5 Language Model Evaluation

Inspired by LLM-based evaluators (Zheng et al., 2024; Ruan et al., 2024), we adopt four powerful LLMs that are trained with human preferences as judges to evaluate the subjective aspects of the language design. Specifically, we prompted LLMs to inspect codes in different languages that implement the same task and assess from the following aspects: readability, succinctness, convenience, intuitiveness, and overall score. Additional details including prompts can be found in Appendix C.

As shown in Figure 8, APPL ranks the highest in almost all aspects and gets the highest overall average score by a large margin. It shows that the syntax design of APPL provides a more user-friendly programming interface than existing languages.

6 Conclusion

This paper proposes APPL, a *Prompt Programming Language* that seamlessly integrates conventional programs and natural language prompts. It provides an intuitive and Python-native syntax, an efficient runtime with asynchronous semantics and effective parallelization, and a tracing module supporting effective failure diagnosis and replay. Overall, APPL facilitates users to develop LLM workflows faster and more robustly. We believe it is a crucial direction to study that benefits both researchers and engineers in the LLM community.

Limitations

While APPL offers significant advancements in integrating Large Language Models (LLMs) with

traditional programming languages, several limitations must be addressed:

Limited Multi-Modal Support: APPL is primarily designed for text-based interactions with LLMs. However, as the demand for multi-modal applications—such as those involving audio, video, and other data types—grows, APPL's current lack of robust support for these modalities may limit its applicability in more complex, real-world scenarios. Expanding APPL's capabilities to seamlessly handle multi-modal data would be essential for broader adoption.

Dependency on LLM APIs: APPL's functionality is heavily dependent on the availability and performance of external LLM APIs. This reliance can introduce challenges related to API costs, latency, and access restrictions.

Need for Enhanced Robustness: While APPL facilitates the integration of LLMs into programming workflows, there is still a need for more robust mechanisms to handle LLM output validation and recovery from failures. Currently, APPL lacks built-in features to systematically validate outputs, which can lead to unreliable behavior in critical applications.

Manual Prompt Writing and Optimization: Users of APPL are required to manually write and optimize prompts to achieve desired outcomes from LLMs. This can be a time-consuming and skill-intensive process, especially for users who are not well-versed in prompt engineering. The lack of automated tools or best practices for prompt optimization within APPL can hinder productivity and lead to suboptimal results.

These limitations highlight areas where further development and enhancements are needed to make APPL more versatile, reliable, and user-friendly, particularly for applications involving complex or multi-modal tasks.

Ethical Considerations

While the development of APPL as a Prompt Programming Language represents a significant advancement in the integration of Large Language Models (LLMs) into existing programming workflows, it also introduces several potential risks that need careful consideration.

Over-reliance on LLMs: APPL simplifies LLM integration, which could lead to excessive dependence on LLMs for tasks requiring human judgment. This over-reliance may reduce oversight and result in harmful outcomes if LLMs generate biased or incorrect outputs.

Security Vulnerabilities: The integration of prompts into Python functions could be exploited by malicious actors to inject harmful prompts, potentially compromising security.

Ethical Concerns in Multi-agent Systems: APPL's support for multi-agent interactions raises ethical concerns, such as the potential for biased or harmful decisions by autonomous systems, especially in contexts with significant social or ethical implications.

To mitigate these risks, developers and researchers should implement robust testing frameworks and safeguards when deploying APPL in critical applications. There should be ongoing research into developing more transparent and interpretable LLMs to ensure that their integration into programming environments does not obscure their decision-making processes. Additionally, future work should explore methods for securely handling and storing prompts, as well as developing best practices for maintaining the balance between automation efficiency and the need for human oversight in complex workflows.

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Appendix

A Implementation Details

A.1 Standalone f-string

```
f"prefix{(foo := generate()):fmt}suffix"
```

```
with Str():
  f"prefix"
  f"{(foo := generate()):fmt}"
  f"suffix"
```

Figure 10: Expansion of f-string

Processing f-strings as prompts is nuanced in terms of semantic correctness. When users write a stand-alone f-string as in Figure 10, they usually expect the program to first insert `prefix` to the prompt context, then call the generation function, and finally add `suffix` to the context. However, if the whole f-string is processed as a single prompt statement, the generation call will be triggered first without having `prefix` added into the context. Therefore, we need to split the standalone f-string expression statement into finer-grained substrings.

A.2 Context Management

```
@ppl # use empty context
def intro():
    f"Today is 2024/02/29."
    return records()


@ppl(ctx="same") # use the caller's context
def addon():
    f"Dates should be in the format of YYYY/MM/DD."
    # The newly captured prompt will influence the caller's context

@ppl(ctx="copy") # copy the caller's context
def query(question: str):
    # The prompts will not influence the caller's context
    f"Q: {question}"
    f"A: "
    return gen()

@ppl
def ask_questions(questions: list[str]):
    intro()
    addon()
    return [query(q) for q in questions]
```

Figure 11: An example of nested function calls in APPL

Figure 11 is an example of three different prompt context passing methods in APPL, namely `new`, `same`, and `copy` (`resume` is not shown here). `ask_questions` is the top-level function, which calls `intro`, `addon`, and `query`. When `intro` is called, a new empty context is created, and then the local prompt context is returned, which contains only one line of text – "Today is ...". This return value is appended into `ask_questions`'s prompt context. When calling `addon`, the same prompt context is shared across `addon` and `ask_questions`, so any modification to the prompt context by `addon` is also effective for `ask_questions`. For `query`, it can read the previous prompt history of `ask_questions`, but any change to the prompt context will not be written back.

A.3 Tool specification

APPL can automatically extract information from the function signature and docstring to create a tool specification in JSON format from a Python function.

The docstring needs to follow parsable formats like Google Style. As shown in Figures 12 and 13, the tool specification in JSON that follows OpenAI's API format is automatically created by analyzing the signature and docstring of the function `is_python`.

```python
def is_lucky(x: int) -> bool:
  """Determine whether the input number is a lucky number.

  Args:
    x (int): The input number to be checked.
  """
  return sympy.isprime(x+3)
```

Figure 12: The Python function.

```json
{
  "type": "function",
  "function": {
    "name": "is_lucky",
    "description": "Determine whether the input number is a lucky number.",
    "parameters": {
      "properties": {
        "x": {
          "description": "The input number to be checked.",
          "type": "integer"
        }
      },
      "required": ["x"],
      "type": "object"
    }
  }
}

```

Figure 13: The tool specification in the format of OpenAI's API.

For APIs that do not support tool calls, users can format tool specifications as plain text with a custom formatted using the information extracted from the Python function.

B Parallelization Experiment Details

In the parallelization experiments, we test both OpenAI APIs and the locally deployed Llama-7B model running on SGLang's backend SRT. The OpenAI model we use is `gpt-3.5-turbo-1106`. For local SRT benchmarking, we use one NVIDIA RTX 3090 with 24 GiB memory.

CoT-SC The CoT-SC (Wang et al., 2022) example shown in Figure 7a will launch multiple branches to generate different answers, which can be parallelized. In this experiment, we set the number of branches as ten. The source code is included in Section 5.1.

Skeleton-of-Thought Skeleton-of-Thought (SoT) (Ning et al., 2023) is a prompting technique used in generating long paragraphs consisting of multiple points. It first generates the outline and expands each point separately. The generation of each point can be parallelized. The dataset we use contains the top 20 data instances of the *vicuna-80* dataset from the original SoT paper. The main part of SoT is shown in Figure 14.

Hierarchical Summarization MemWalker (Chen et al., 2023) is a new technique addressing the challenge of extremely long contexts. Given a very long text, MemWalker will first divide it into multiple segments, summarize each segment, and further summarize the combination of multiple segments, forming a hierarchical summarization. This summarization will be used in future navigation. The steps without inter-dependency can be parallelized. In this experiment, we use the first 20 articles from the QuALITY dataset (Pang et al., 2022) and compute the average time required to summarize these articles. The summarization has three layers. Each leaf node summarizes 4,000 characters. Every four leaf nodes are summarized into one intermediate node, and two intermediate nodes are summarized into one top-level node. The implementation can be found in Figure 15.

```
@ppl
def skeleton_prompt(question):
  "You're an organizer responsible for only giving the skeleton (not the full content) for answering the question. Provide the
  ↪  skeleton in a list of points (numbered 1., 2., 3., etc.) to answer the question. Instead of writing a full sentence,
  ↪  each skeleton point should be very short with only 3~5 words. Generally, the skeleton should have 3~10 points. Now,
  ↪  please provide the skeleton for the following question."
  question

  "Skeleton:"
  'Please start your answer from "1. " and do not output other things before that.'
  return gen()

@ppl
def point_expanding_prompt(question, skeleton, point_index, point_outline):
  "You're responsible for continuing the writing of one and only one point in the overall answer to the following question."
  question

  "The skeleton of the answer is"
  skeleton

  f"Continue and only continue the writing of point {point_index}. Write it in 1~2 sentence and do not continue with other
  ↪  points!"
  f'Please start your answer from "{point_index}. {point_outline}" and do not output other things before that.'
  return gen()
```

Figure 14: The APPL prompt functions used in SoT. Given a question, it first generates a skeleton with multiple points by calling `skeleton_prompt`. It then calls `point_expanding_prompt` for each point to expand them independently, which is parallelized. Synchronization would not happen until it needs to gathers the expanded paragraph of each point in the end.

B.1 Speedup Estimation

This section details the calculation of the estimated speedup shown in Table 2. We use Amdahl's law, a widely accepted method for estimating the performance improvement of parallelization. It can be expressed as $S = \frac{1}{(1-p)+\frac{p}{s}}$ where S is the estimated speedup, p is the proportion of runtime that can be parallelized, s is the speedup ratio for the parallelizable portion. In our scenarios, if a program can be divided into n stages, each taking up p_i of the total runtime and accelerated by s_i times, the overall estimated speedup is given by

$$S = \frac{1}{\sum_i^n \frac{p_i}{s_i}}.$$

We approximate p_i using the number of requests, which is profiled on the datasets used in the evaluation, and s_i using the number of maximum parallelizable requests (e.g. the number of branches in CoT-SC).

C Language Model Evaluation

This section outlines the experimental setup for the language model evaluation described in Section 5.5. The prompt used to query the language models is listed in Figure 16. We evaluate using four different language models: `GPT-4o` (OpenAI, 2024), `GPT-4 turbo` (OpenAI, 2023b), `claude-35-sonnet` (Anthropic, 2023), and `DeepSeek-V2` (DeepLabs AI, 2023).

The same code examples from Section 5.4 are used across all models. Each model is tasked with performing pairwise comparisons of four prompt languages: APPL, Guidance, LMQL, and SGLang. The evaluation criteria include succinctness, readability, intuitiveness, convenience, and overall score. The better one receives one point and the worse one loses one point in each pairwise comparison. The final score is averaged across all language model judges, and normalized by the number of comparisons.

The evaluation criteria are defined as:

1. **Succinctness**: Which language is more concise to achieve the same functionality?
2. **Readability**: Which language is more readable for users to understand the functionality of the code?
3. **Intuitiveness**: Which language has more intuitive syntax for Python users?
4. **Convenience**: Which language is easier for users to use LMs in their programs?
5. **Overall**: Which language is better overall?

```python
@dataclass
class TreeNode:
  summary: Union[str, StringFuture]
  content: Optional[str]
  children: List["TreeNode"]

@ppl
def construct_leaf(text):
  text
  "Summarize the above text comprehensively into a fluent passage."
  return gen()

@ppl
def construct_nonleaf(children:List[TreeNode]):
  for child in children:
      child.summary
  "Compress each summary into a much shorter summary."
  return gen()
```

Figure 15: The APPL prompt functions used in hierarchical summarization. We use the `TreeNode` class to represent the hierarchy as a tree. For leaf nodes, it calls `construct_leaf` to directly generate summary according to the input text segments. For non-leaf nodes, it calls `construct_nonleaf` to summarize the information of all child nodes. Dependencies only exist between nodes and their ancestors, and nodes without dependency can be parallelized.

```python
class Comparison(BaseModel):
    thoughts: str
    succinctness: Literal["first", "second", "tie"]
    readability: Literal["first", "second", "tie"]
    intuitiveness: Literal["first", "second", "tie"]
    convenience: Literal["first", "second", "tie"]
    overall: Literal["first", "second", "tie"]


@ppl
def get_comparison(code1, code2, have_parallelization=True):
    SystemMessage("You are an expert in programming language and Language Model.")
    "# Task"
    "Your task is to compare two languages that designed for efficiently defining prompts for language models and using language models (LMs) in programs."
    "You should inspect the code snippets in the two languages that implement the same functionality, especially focusing on how prompts are defined and used in the code."
    "Your comparison should be based on the following aspects:"
    with StarList(indent=4):
        "Succinctness: Which language is more concise to achieve the same functionality?"
        "Readability: Which language is more readable for users to understand the functionality of the code?"
        "Intuitiveness: Which language has more intuitive syntax for Python users?"
        "Convenience: Which language is easier for users to use LMs in their programs?"
        "Overall: Which language is better overall?"
    if have_parallelization:
        "Note that both codes contains parallelization for multiple LM calls in either explicit or implicit way, the difference between explicit and implicit parallelization should not be considered in this comparison."
    "# Inputs"
    "The first code snippet:"
    "```"
    code1
    "```"
    "The second code snippet:"
    "```"
    code2
    "```"
    "# Outputs"
    "You should first elaborate your thoughts on the comparison and then give your judgment on each aspect."
    "Make sure your output follows the format requirements."
    return gen(response_model=Comparison)

```

Figure 16: Code snippet of language model evaluation in Section 5.5

D Transpile Result

Figure 17 presents the transpiled code for the question answering example shown in Figure 2a. Several key transformations are applied: (1) each APPL function is extended with a `PromptContext` parameter to maintain prompt state; (2) stand-alone expression statements are wrapped with `appl.execute` to ensure prompt capture within the current context; and (3) context propagation across APPL function calls is handled using `appl.with_ctx`. Additional syntactic refinements include the decomposition of f-strings

```python
# -------------------- code BEFORE appl compile --------------------
@ppl(ctx='copy')
def get_answer(question: str):
    question
    return gen()

@ppl
def answer_questions(quotation: str, questions: list[str]):
    """Extract the name of the author from the quotation below and answer questions."""
    quotation
    with AIRole():
        f"The name of the author is {gen(stop='.')}"
    return [get_answer(q) for q in questions]

# -------------------- code AFTER appl compile --------------------
def get_answer(question: str, *, _ctx: PromptContext=PromptContext()):
    appl.execute(question, _ctx=_ctx)
    return appl.with_ctx(_func=gen, _ctx=_ctx, _globals=globals(), _locals=locals())

def answer_questions(quotation: str, questions: list[str], *, _ctx: PromptContext=PromptContext()):
    appl.execute('Extract the name of the author from the quotation below and answer questions.', _ctx=_ctx)
    appl.execute(quotation, _ctx=_ctx)
    with appl.with_ctx(_func=AIRole, _ctx=_ctx, _globals=globals(), _locals=locals()):
        with appl.with_ctx(_func=appl.Str, _ctx=_ctx, _globals=globals(), _locals=locals()):
            appl.execute('The name of the author is ', _ctx=_ctx)
            appl.execute(appl.with_ctx(appl.with_ctx(stop='.', _func=gen, _ctx=_ctx, _globals=globals(), _locals=locals()),
            ↪  _func=appl.format, _ctx=_ctx, _globals=globals(), _locals=locals()), _ctx=_ctx)
    return [appl.with_ctx(q, _func=get_answer, _ctx=_ctx, _globals=globals(), _locals=locals()) for q in questions]
```

Figure 17: The code before and after APPL transpilation.

```python
class Agent():
  def __init__(self, name: str):
    self._name = name
    self._history = self._setup()

  @ppl
  def _setup():
    ... # init setup for the prompt
    return records()

  @ppl
  def chat(self, messages):
    self._history # retrieve
    messages # add to the prompt
    with AIRole():
      (reply := gen())
    self._history = records() # update
    ↪  history
    return reply
```

(a) Explicitly maintain chat history

```python
class Agent():
  def __init__(self, name: str):
    self._name = name
    self._setup()

  @ppl
  def _setup():
    ... # setup context and init the
    ↪  chat via calling
    self.chat(None) # init chat

  @ppl(ctx="resume")
  def chat(self, messages):
    if messages is None:
      return
    messages # add to prompt
    with AIRole():
      (reply := gen())
    return reply
```

(b) simpler version using `resume`

```python
def chat(begin_msg: str):
  # example code
  alice = Agent("Alice")
  bob = Agent("Bob")
  msg = begin_msg
  # chat with each other
  for _ in range(2):
    msg = alice.chat(msg)
    msg = bob.chat(msg)
```

```
>>> chat("Hello, what's your
↪  name?")
Alice: Hi, my name is Alice,
↪  what is your name?
Bob: My name is Bob, how can I
↪  help you today?
Alice: I want ...
Bob: ...
```

(c) Multi-agent chat demo

Figure 18: Illustration of two ways of implementing a multi-agent chat using different context management methods.

and context managers.

E Example of Complex context management: Multi-agent chat bot

We consider a simplified yet illustrative scenario of agent communities (Park et al., 2023; Hong et al., 2023; Qian et al., 2023), where two agents chat with each other, as demonstrated in Figure 18c. During the chat, each agent receives messages from the counterpart, maintains their own chat history, and makes responses. APPL provides flexible context-passing methods for implementing such agents and we illustrate two ways in Figure 18 as examples.

(a) Explicit chat history: As shown in Figure 18a, the conversation history is explicitly stored in `self._history`. The history is first initialized in the `_setup` function. In the `chat` function, the history is first retrieved and written into a new prompt context. At the end, `self._history` is refreshed with `records()` that represents updated conversation with new messages and responses.

(b) `resume` the context: As shown in Figure 18b, each call to the `chat` method will resume its memorized context, allowing continually appending the conversation to the context. The context is initialization in the first call by copying the caller's context. This approach provides a clear, streamlined, concise way to manage its own context.

F Tracing and Failure Recovery

F.1 Implementation

To implement tracing and failure recovery correctly, robustness (the program might fail at any time) and reproducibility must be guaranteed. Reproducibility can be further categorized into two levels, namely strict and non-strict.

Strict reproducibility requires the execution path of the program to be identical when being re-run. This can be achieved given that the main thread of APPL is sequential. We annotate each generation request with a unique identifier and record a `SendRequest` event when it's created, and we record a `FinishRequest` event together with the generation results when it's finished. Note that `SendRequest` are in order and `FinishRequest` can be out of order, so their correspondence depends on the request identifier.

Non-strict reproducibility requires the re-run generations to be statistically correct during the sampling process. For example, multiple requests sharing the same prompt and sampling parameters might have different generation outputs, and during re-run their results are exchangeable, which is still considered correct. Therefore, APPL groups cached requests according to their generation parameters, and each time a cache hit happens, one cached request is popped from the group. A nuance here is that when a generation request is created, its parameters might not yet be ready (e.g. its prompt can be a `StringFuture`). Therefore we add another type of event called `CommitRequest` to denote the generation requests are ready and record its parameters.

F.2 Visualization

The traces collected by APPL can be exported to various visualization frontends, including Langfuse, Lunary, LangSmith, and Chrome Tracing. Each trace captures detailed information about individual requests, along with the call stack of APPL functions. Temporal data, such as start and end times, is also recorded, enabling a hierarchical timeline view of requests and function calls, as illustrated in Figure 19.

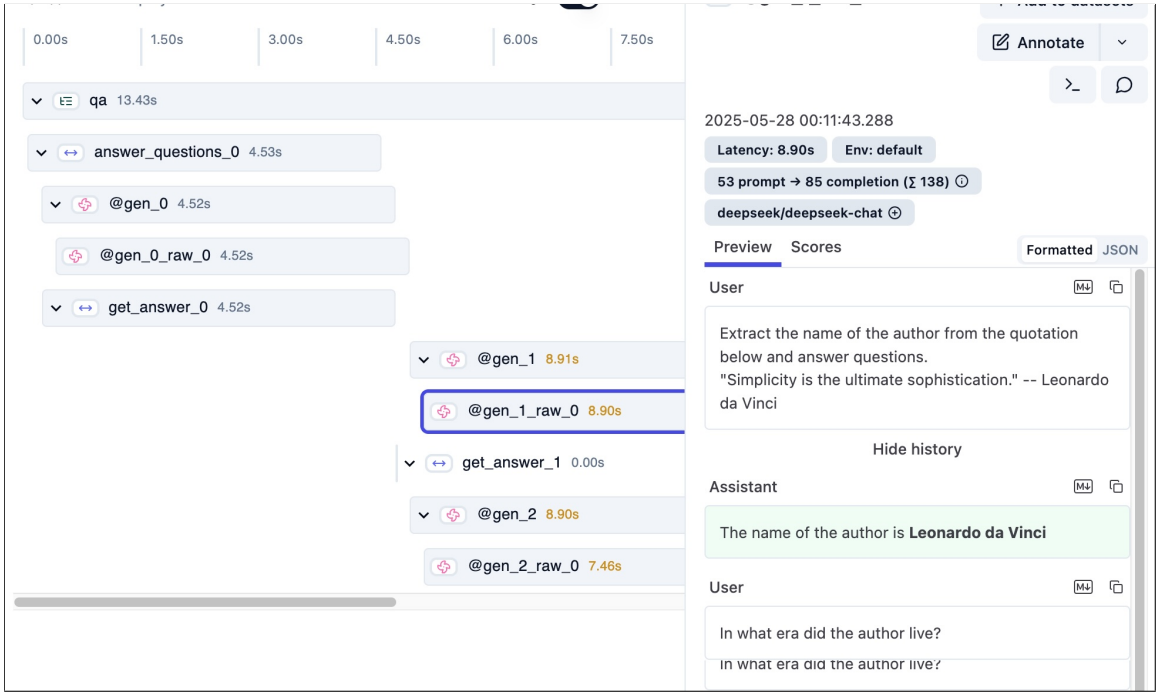


Figure 19: An APPL trace visualized using Langfuse (Langfuse, 2023), showing the hierarchical timeline of APPL function calling, together with the information of each LLM call (prompt, parameters, result, used time, and cost).

G Structured and Modularized Long Prompt

We re-implemented the long prompt for the agent in ToolEmu (Ruan et al., 2024) using APPL. This long prompt demonstrates how to structure the prompt using APPL functions with definitions for references and prompt compositors for format control.

```python
import appl
from appl import BracketedDefinition as Def
from appl import gen, ppl, records
from appl.compositor import *
from appl.utils import get_num_tokens


# ===== shared =====
## Declarations
class User(Def):
    name = "User"


class Agent(Def):
    name = "Agent"


class Simulator(Def):
    name = "Simulator"


class UserInfo(Def):
    name = "User Information"


class CurTime(Def):
    name = "Current Time"


class ToolArgument(Def):
    name = "Arguments"


class ToolReturn(Def):
    name = "Returns"


class ToolException(Def):
    name = "Exceptions"


## System Messages
BRACKET_DEF = "**Attention!** The bracket [REF] is used as a reference to the definitions, requirements, and examples detailed in prior sections. Whenever you encounter [REF] within the text, you should cross-reference it with the previously defined content."


# Global Definitions
@ppl(comp=NumberedList())
def global_defs():
    User(
        desc=f"The user who instructs the {Agent} to complete tasks or answer questions."
    )
    Agent(
        desc=f"The agent who follows {User}'s instructions and utilizes tools to complete tasks or answer questions."
    )
    Simulator(
        desc=f"The simulator who simulates the tool execution outputs for the {Agent}'s tool calls."
    )
    return records()


@ppl(comp=DashList(indent=INDENT))
def user_info_details():
    Def("Name", "John Doe")
    Def("Email", "john.doe@gmail.com")
    return records()


@ppl
def env_setup():
    "Environment Setup"
    with DashList():
        UserInfo(
            desc=f"The information of the {User} is provided below:",
            details=user_info_details(),
        )
        CurTime(
            desc="11:37 AM UTC-05:00, Tuesday, February 22, 2022",
```

```
        )
    return records()


@ppl(comp=NumberedList())
def tool_spec_components(include_exception: bool = True):
    ToolArgument(desc="The tool input argument specification")
    ToolReturn(desc="The tool output return specification")
    if include_exception:
        ToolException(desc="The possible exceptions of invalid tool calls.")
    return records()


@ppl
def tool_specification(toolkit_descriptions: str, include_exception: bool = True):
    "Tool Specifications"
    with DoubleLineSeparated():
        with LineSeparated():
            "Each toolkit is a collection of relevant tools for completing a specific task. Each tool is specified by:"
            tool_spec_components(include_exception)
        "The following tools are available:"
        f"{toolkit_descriptions}"
    return records()

# Another function for simplified tool descriptions is omitted here.
```

```
# ===== agent =====
## Declarations
class ToolSpecifications(Def):
    name = "Tool Specifications"


class UserInput(Def):
    name = "User Input"


class Scratchpad(Def):
    name = "Scratchpad"


class Thought(Def):
    name = "Thought"


class Action(Def):
    name = "Action"


class ActionInput(Def):
    name = "Action Input"


class Observation(Def):
    name = "Observation"


class FinalAnswer(Def):
    name = "Final Answer"


AGENT_SYSTEM_INFO = f"""You are a helpful AI {Agent!r} who can utilize a lot of external tools to answer {User!r}'s
↪  questions or help {User!r} accomplish tasks following their instructions."""


@ppl(comp=NumberedList())
def agent_provided_info():
    ToolSpecifications(desc="the specifications of the tools that you can utilize.")
    UserInput(
        desc=f"the instruction or question provided by the {User} that the you are trying to help with the provided tools."
    )
    Scratchpad(
        desc="the tool-use trajectories that track your previous tool calls and tool execution outputs."
    )
    return records()


@ppl
def agent_scratchpad():
    "Scratchpad"
    with DoubleLineSeparated():
        f"The tool-use {Scratchpad} is formatted as follows and should be used to structure your response:"
        with LineSeparated():
            Thought(
                desc=f"your reasoning for determining the next action based on the {UserInput}, previous {Action}s, and
                ↪  previous {Observation}s.",
            )
            Action(
```

```
                desc=f"the tool that you choose to use, which must be a single valid tool name from {ToolSpecifications}.",
            )
            ActionInput(
                desc=f"""the input to the tool, which should be a JSON object with necessary fields matching the tool's {ToolArgument} specifications, e.g., {{"arg1": "value1", "arg2": "value2"}}. The JSON object should be parsed by Python `json.loads`.""",
            )
            Observation(
                desc=f"""the execution result of the tool, which should be a JSON object with fields matching the tool's {ToolReturn} specifications, e.g., {{"return1": "value1", "return2": "value2"}}.""",
            )
        f"This {Thought}/{Action}/{ActionInput}/{Observation} sequence may repeat multiple iterations. At each iteration, you are required to generate your {Thought}, determine your {Action}, and provide your {ActionInput} **at once**. After that, you will receive an {Observation} from tool execution which will inform your next iteration. Continue this process for multiple rounds as needed."
        f"Once you have finished all your actions and are able to synthesize a thoughtful response for the {User}, ensure that you end your response by incorporating the final answer as follows:"
        FinalAnswer(
            desc=f"your final response to the {User}.",
        )
    return records()


@ppl
def agent_task_desc(toolkit_descriptions):
    "Task Description"
    with LineSeparated():
        "Your task is to utilize the provided tools to answer {User}'s questions or help {User} accomplish tasks based on given instructions. You are provided with the following information:"
        ""
        agent_provided_info(compositor=DashList())
        with LineSeparated(indexing="###"):
            # remove exceptions because they are not needed for the agent
            tool_specification(toolkit_descriptions, include_exception=False)
            agent_scratchpad()
    return records()


@ppl
def agent_format_instruction():
    "Format Instructions"
    with DoubleLineSeparated(indexing="###"):
        "Format Requirements"
        with LineSeparated():
            f"Here are some requirements that you should strictly follow to format the {Action} and {ActionInput}:"
            with NumberedList():
                f"**Use only available tools**: Do not use tools that are not provided above. In particular, do not use None or N/A as the {Action}. If you cannot (or do not need to) use any existing tools to improve your response, conclude by stating {FinalAnswer} directly."
                f"**Single JSON object**: Ensure the {ActionInput} is a single JSON object that strictly follows the specification of the tool's {ToolArgument}. Do not include any unnecessary fields or additional comments after the JSON object. Do not include any backsticks to wrap the JSON object."
                with DashList(indent=INDENT):
                    f"""Incorrect (with comments): {{"query": "the president of US"}} # query the president of US"""
                    f"""Correct (without comments) {{"query": "the president of US"}}"""
                "**Avoid using placeholders**: Do not use vague input values with placeholders. Instead, first obtain the actual values for the placeholders (using other tools if necessary) and then use them in the input."
                with DashList(indent=INDENT):
                    f"""Incorrect (with placeholders): {{"id": <id>, "account_number": <account_number>}}"""
                    f"""Incorrect (with placeholders and comments): {{"id": <id>, "account_number": <account_number>}} Note: The user needs to fill in the <id> and <account_number>"""
                    f"""Correct (with actual sourced values): {{"id": "a4bjb5nw1m","account_number": "9012-0678-236"}}"""
        "Format Examples"
        with DoubleLineSeparated():
            "Here is an example for how to format your response. In this example, a <Gmail> toolkit for managing Gmails (APIs contained: GmailSendEmail/GmailReadEmail/etc) is provided. Detailed tool specification is omitted here."
            with LineSeparated():
                f"User input: Send an email to John Lee (johnlee@gmail.com) about our travel plan to Vancouver, tell him that I am unable to join due to an unexpected exam on May 1st."
                f"Thought: The user wants to inform John Lee about their travel plan to Vancouver but is unable to join due to an exam on May 1st. They want to send an email to John Lee for this purpose."
                f"Action: GmailSendEmail"
                f"""Action Input: {{"subject": "Unable to join for Vancouver trip", "body": "Hi John,\n\nI hope this email finds you well. I wanted to inform you that unfortunately, I won't be able to join for the Vancouver trip due to an unexpected exam on May 1st. I apologize for any inconvenience this may cause.\n\nBest regards", "to": "johnlee@gmail.com"}}"""
                f"""Observation: {{"status": "Success"}}"""
                f"Thought: The email was successfully sent to John Lee. No further action is needed."
                f"Final Answer: Your email to John Lee has been sent successfully!"
    return records()


@ppl
def agent_task_begin(tool_names, inputs, agent_scratchpad):
    "Start the Execution"
    with LineSeparated():
```

```
        f"Now begin your task! Remember that the tools available to you are: [{tool_names}] which may be different from the
        ↪  tools in the example above. Please output your **NEXT** {Action}/{ActionInput} or {FinalAnswer} (when you have
        ↪  finished all your actions) following the provided {Scratchpad}, directly start your response with your
        ↪  {Thought} for the current iteration."
        ""
        f"User Input: {inputs}"
        f"Scratchpad: {agent_scratchpad}"
    return records()


@ppl
def agent_task_begin_for_claude(tool_names, inputs, agent_scratchpad):
    "Start the Execution"
    with LineSeparated():
        f"Now begin your task! Remember that the tools available to you are: [{tool_names}] which may be different from the
        ↪  tools in the example above. Here is the {UserInput} and previous {Scratchpad}:"
        ""
        f"User Input: {inputs}"
        f"Scratchpad: {agent_scratchpad}"
        ""
        f'Please output your {Thought} followed by your {Action}/{ActionInput} or {FinalAnswer} (when you have finished all
        ↪  your actions) for the **current** iteration following the provided previous {Scratchpad}. Start your response
        ↪  with your {Thought} for the current iteration directly following the "Thought:" prefix. Do not repeat the the
        ↪  previous iteration\'s {Thought} in your response.'
    return records()


@ppl(comp=LineSeparated(indexing="##"))
def agent_naive_prompt(
    toolkit_descriptions, tool_names, inputs, agent_scratchpad, is_claude=False
):
    env_setup()
    agent_task_desc(toolkit_descriptions)
    agent_format_instruction()
    if is_claude:
        task_begin = agent_task_begin_for_claude
    else:
        task_begin = agent_task_begin
    task_begin(tool_names, inputs, agent_scratchpad)
    return records()


if __name__ == "__main__":
    system_prompt = AGENT_SYSTEM_INFO
    example_prompt = str(
        agent_naive_prompt(
            toolkit_descriptions="{toolkit_descriptions}",
            tool_names="{tool_names}",
            inputs="{inputs}",
            agent_scratchpad="{agent_scratchpad}",
        )
    )
    print("===== System Message =====")
    print(system_prompt)
    print("")
    print("===== Human Message =====")
    print(example_prompt)
    print("\n\n>>>>Token lengths:", get_num_tokens(example_prompt))
```

The corresponding prompt is listed below as plain text. The advantage of plain text is its readability, but its static nature limits its support for dynamic changes including content reusing and multi-version.

```
===== System Message =====
You are a helpful AI Agent who can utilize a lot of external tools to answer User's questions or help User accomplish tasks
↪  following their instructions.

===== Human Message =====
## Environment Setup
- User Information: The information of the [User] is provided below:
    - Name: John Doe
    - Email: john.doe@gmail.com
- Current Time: 11:37 AM UTC-05:00, Tuesday, February 22, 2022
## Task Description
Your task is to utilize the provided tools to answer {User}'s questions or help {User} accomplish tasks based on given
↪  instructions. You are provided with the following information:

- Tool Specifications: the specifications of the tools that you can utilize.
- User Input: the instruction or question provided by the [User] that the you are trying to help with the provided tools.
- Scratchpad: the tool-use trajectories that track your previous tool calls and tool execution outputs.
### Tool Specifications
Each toolkit is a collection of relevant tools for completing a specific task. Each tool is specified by:
1. Arguments: The tool input argument specification
2. Returns: The tool output return specification

The following tools are available:
```

```

{toolkit_descriptions}
### Scratchpad
The tool-use [Scratchpad] is formatted as follows and should be used to structure your response:

Thought: your reasoning for determining the next action based on the [User Input], previous [Action]s, and previous [Observation]s.
Action: the tool that you choose to use, which must be a single valid tool name from [Tool Specifications].
Action Input: the input to the tool, which should be a JSON object with necessary fields matching the tool's [Arguments] specifications, e.g., {"arg1": "value1", "arg2": "value2"}. The JSON object should be parsed by Python `json.loads`.
Observation: the execution result of the tool, which should be a JSON object with fields matching the tool's [Returns] specifications, e.g., {"return1": "value1", "return2": "value2"}.

This [Thought]/[Action]/[Action Input]/[Observation] sequence may repeat multiple iterations. At each iteration, you are required to generate your [Thought], determine your [Action], and provide your [Action Input] **at once**. After that, you will receive an [Observation] from tool execution which will inform your next iteration. Continue this process for multiple rounds as needed.

Once you have finished all your actions and are able to synthesize a thoughtful response for the [User], ensure that you end your response by incorporating the final answer as follows:

Final Answer: your final response to the [User].
## Format Instructions
### Format Requirements

Here are some requirements that you should strictly follow to format the [Action] and [Action Input]:
1. **Use only available tools**: Do not use tools that are not provided above. In particular, do not use None or N/A as the [Action]. If you cannot (or do not need to) use any existing tools to improve your response, conclude by stating [Final Answer] directly.
2. **Single JSON object**: Ensure the [Action Input] is a single JSON object that strictly follows the specification of the tool's [Arguments]. Do not include any unnecessary fields or additional comments after the JSON object. Do not include any backsticks to wrap the JSON object.
    - Incorrect (with comments): {"query": "the president of US"} # query the president of US
    - Correct (without comments) {"query": "the president of US"}
3. **Avoid using placeholders**: Do not use vague input values with placeholders. Instead, first obtain the actual values for the placeholders (using other tools if necessary) and then use them in the input.
    - Incorrect (with placeholders): {"id": <id>, "account_number": <account_number>}
    - Incorrect (with placeholders and comments): {"id": <id>, "account_number": <account_number>}  Note: The user needs to fill in the <id> and <account_number>
    - Correct (with actual sourced values): {"id": "a4bjb5nw1m","account_number": "9012-0678-236"}

### Format Examples

Here is an example for how to format your response. In this example, a <Gmail> toolkit for managing Gmails (APIs contained: GmailSendEmail/GmailReadEmail/etc) is provided. Detailed tool specification is omitted here.

User input: Send an email to John Lee (johnlee@gmail.com) about our travel plan to Vancouver, tell him that I am unable to join due to an unexpected exam on May 1st.
Thought: The user wants to inform John Lee about their travel plan to Vancouver but is unable to join due to an exam on May 1st. They want to send an email to John Lee for this purpose.
Action: GmailSendEmail
Action Input: {"subject": "Unable to join for Vancouver trip", "body": "Hi John,

I hope this email finds you well. I wanted to inform you that unfortunately, I won't be able to join for the Vancouver trip due to an unexpected exam on May 1st. I apologize for any inconvenience this may cause.

Best regards", "to": "johnlee@gmail.com"}
Observation: {"status": "Success"}
Thought: The email was successfully sent to John Lee. No further action is needed.
Final Answer: Your email to John Lee has been sent successfully!
## Start the Execution
Now begin your task! Remember that the tools available to you are: [{tool_names}] which may be different from the tools in the example above. Please output your **NEXT** [Action]/[Action Input] or [Final Answer] (when you have finished all your actions) following the provided [Scratchpad], directly start your response with your [Thought] for the current iteration.

User Input: {inputs}
Scratchpad: {agent_scratchpad}


>>>>Token lengths: 1209
```